COMMERCIAL ITEM DESCRIPTION

LENS, OPHTHALMIC, Glass

The General Services Administration has authorized the use of this commercial item description in preference to Military Specification MIL-L-16991C.

This commercial item description covers the uncut glass ophthalmic lenses, dress type for visual deficiency correction.

Salient characteristics:

Lenses shall be classified to the following types, classes, styles, sizes and power required, as specified:

Type
I. Lens blank, semifinished
II. Lens, finished, uncut

Class
a. Bifocal
b. Single vision
c. Neutral absorption, N-21

Style
1. Spheres
2. Plano minus cylinders
3. Plus on minus compounds
4. Minus on minus compounds
5. One-piece raised ledge
6. 25mm straight top

Size. The lens diameter under this specification shall be 65 mm ± 1 mm round or 65 mm X 60 mm ± 0.5 mm rectangular with rounded corners. Other shapes are acceptable provided equivalent size is met.

All glass lenses shall be capable to be hardened by chemical or dry heat tempering process after edging to withstand the impact resistance test requirement as specified in ANSI Z80.1-1979, Recommendations for Prescription Ophthalmic Lenses. When requested, the contractor shall demonstrate the impact resistance test for the satisfaction of the contracting officer's representative.

The finished lens shall be a member of an integrated series of ophthalmic lenses having surface curvatures computed to conform to the corrected curve concept.

AMSC N/A FSC 6540

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:15Z
Check the source to verify that this is the current version before use.

Refractive power tolerance for the finished lens shall be as follows:

Sphere	0.00 thru 6.50 diopter is ± 0.13D
	Greater than +6.50 diopter or less than -6.50 diopter is ±2%
Cylinder	-0.25 thru -2.00 diopter is ±0.13D
	-2.12 thru -4.50 diopter is ±0.15D
	Less than -4.50 diopter is ±4%

The base curve on minus lens shall have certain dioptric power commensurate with the curve established by the major lens manufacturers. Flat base curve is not acceptable. Shall be to specified axial back vertex power within a tolerance of plus or minus 0.06 diopter. Center thickness of minus lens shall not be less than 2.2 mm. The corrective curve shall be generated on the inside of the lens.

The semifinished lens shall have the base curve located on convex side only.

The white glass lens shall be made of high grade white ophthalmic quality crown glass with an index of refraction of 1.5230 ± 0.0015. The lens shall be free of bubbles, inclusions, striae and cloudiness. When exposed to sunlight, it shall remain clear and free of color or shade.

The neutral absorption lens shall be neutral gray density commonly referred as "gray No. 3", daylight transmission with dominant wavelength between 570 to 600 millimicrons range shall be 20 ± 1 percent, near infra-red transmission shall be 20 percent maximum and erythermal ultra-violet transmission shall be 0.2 percent maximum, each to be determined on a glass thickness of 2.2 ± 0.2 mm.

The 25 mm straight top bifocal lens blank shall have the major blank made of white ophthalmic crown glass with an index of refraction of 1.523 ± .0015. The segment shall also be white, made of barium glass. The dispersion of the segment glass, Nu value (V) shall be equal to, or greater than the value defined by the formula:

$$V = \frac{16.7\ (n-1)}{n-1.3744}$$

Where "n" is the index of refraction of the segment glass for the sodium D line.

Segment size shall be 25 mm (+0.75/-0.50) wide by 17-1/2 mm (±0.5) high and decentered 3.00 mm (+ 1.5/1.0) to the right and left of the vertical midline. The distance from the geometric center of the major blank to the top of the segment shall be 3 to 6 mm. The optical center of the segment shall be 5 mm ± 0.5 below the flat top and midway between the lateral edges of the segment. Finished distance power shall be within ± 0.06 diopter of specified power. Segment addition shall be within + 0.06D/-0.09D of specified power.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:15Z
Check the source to verify that this is the current version before use.

The one-piece raised ledge bifocal lens blank shall be made of high grade white ophthalmic quality crown glass with an index of refraction of 1.5230 ± 0.0015. The raised ledge shall be straight across and positioned 3 to 5 mm below the geometrical center of the lens.

Workmanship. The lens shall be free from defects which detract from its appearance or impair its serviceability.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:15Z
Check the source to verify that this is the current version before use.

PACKAGING

Single vision finished lenses or semifinished lenses.

Unit. Pair (PR). One pair of lenses, each lens packaged in a snug fitting envelope, constitutes one unit.

Finished bifocal lenses.

Unit. Pair (PR). One matched pair consisting of one left and one right lens each placed in an envelope and the pair further packaged in a paper box constitutes one unit. As an alternate, the paper box may be omitted.

Semifinished bifocal lenses.

Unit. Pair (PR). Each lens shall be wrapped in tissue and packaged in a snug fitting box. Two boxes shall be taped together to constitute one pair.

Packaging quantities. The number of lenses to be packaged in intermediate and exterior containers shall be as follows:

Single vision finished lens.

Lens power Diopter	Number of units In intermediate package	Number of units In a shipping Container
Plus on minus compound lenses		
+0.25 to +2.25	15	540
+2.50 to +5.00	10	360
+5.25 to +8.00	5	180
Minus cylinder lenses		
-0.25 to -1.25	15	540
-1.50 to -2.50	10	360
-2.75 to -6.00	5	180
Minus on minus compound lenses		
-0.25 to -3.00	10	360
-3.25 to -7.50	5	180

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:15Z
Check the source to verify that this is the current version before use.

Finished bifocal lens

Number of units in intermediate package	Number of units in a shipping container
15	540

Semifinished bifocal lens

Number of units in intermediate package	Number of units in a shipping container
5	180

Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations.

Labeling and marking.

Unit package. Each envelope for single vision finished or semifinished lenses and boxes for finished and semifinished bifocal lenses. Each envelope or box shall be marked with the lens power, lens size, base curve, name or registered trade mark of the contractor, National Stock Number, quantity and unit of issue.

Intermediate package and shipping container. Each intermediate package and shipping container shall be marked as specified in MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:15Z
Check the source to verify that this is the current version before use.

For Information Only. The following National Stock Numbers are covered by this document.

6540-00-104 series	6540-00-965 series
6540-00-164 series	6540-01-088 series
6540-00-177 series	6540-01-089 series
6540-00-180 series	6540-01-090 series
6540-00-518 series	6540-01-091 series
6540-00-528 series	6540-01-095 series
6540-00-782 series	6540-01-102 series
6540-00-926 series	6540-01-158 series

Ordering data. (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order).

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITY:

VA-OSS
PHS
FDA-MPQAS

Project No. 6540-1145

Location: NCR Disk #3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:15Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS. In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 4966 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Personnel Support Center
ATTN DPSC-RSTS
2800 South 20th Street
Philadelphia, Pennsylvania 19101-8419

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:15Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53088	LENS, OPHTHALMIC, GLASS

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street City State ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:15Z
Check the source to verify that this is the current version before use.